

MA203, Module 4, Tutorial 4

- The value of $\int_{0.2}^{2.2} xe^x dx$ by using 3-segment trapezoidal rule is most nearly
(a) 11.672 (b) 11.807 (c) 12.811 (d) 14.633
- The value of $\int_3^{19} f(x) dx$ by using 2-segment Simpson's 1/3 rule is estimated as 702.039. The estimate of the same integral using 4-segment rule Simpson's 1/3 rule most nearly is
(a) $702.039 + \frac{8}{3} [2f(7) - f(11) + 2f(15)]$
(b) $\frac{702.039}{2} + \frac{8}{3} [2f(7) - f(11) + 2f(15)]$
(c) $702.039 + \frac{8}{3} [2f(7) + 2f(15)]$
(d) $\frac{702.039}{2} + \frac{8}{3} [2f(7) + 2f(15)]$
- The highest order of polynomial integrand for which Simpson's 1/3 rule of integration is exact is
(a) first (b) second (c) third (d) fourth
- The function $f(x)$ is tabulated below, for different values of x ,

X:	0	5	10	15	20
f(x):	1.5708	1.5738	1.5825	1.5981	1.6200

Compute the first and second derivatives of $f(x)$ at $x = 0, 3, 15, 18$.

- The value of $\frac{d^2y}{dx^2}$ at $x = 1$, for the table given below is

x	-1	1	2	3
F(x)	7	5	19	51
- On dividing the interval into 10 equal parts and applying Simpson's 1/3rd rule, find the value of the integral $\int_0^5 \frac{dx}{4x+5}$ correct to 4 decimal places.
- Find the value of the integral $\int_0^1 \frac{dx}{1+x}$ using trapezoidal rule with step size $h = \frac{1}{4}$ correct to 4 decimal places.
- Using the following data, find $f'(6.0)$

X	6.0	6.1	6.2	6.3	6.4
f(x)	0.1750	-0.1998	-0.2223	-0.2422	-0.2596

9. Estimate the length of the arc of the curve $3y = x^3$ from $(0, 0)$ to $(1, 1/3)$ using Simpson's $1/3^{\text{rd}}$ rule by taking $h = 0.125$
10. Find the magnitude of the error (correct to two decimal places) in the estimation of the following integral using Simpson's $1/3^{\text{rd}}$ rule. Take step length as 1. $\int_0^4 (x^4 + 10)dx$
11. Using the values of a function $y=f(x)$ given in the following table, find the first two derivatives of $f(x)$ at $x=7$

X	2	3	4	5	6	7
y=f(x)	4	8	15	27	36	42

12. Simpson's rule of integration gives exact result when $f(x)$ is a polynomial of degree
- (a) 1 (b) 2 (c) 3 (d) all of these